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we can represent the state by four variables, and each of them has one of the two values. Let us name the variables farmer, goat, grass and wolf, and their possible values L and R. A value of L means "at the left side". A value of R means "at the right side". Since the boat is always with the farmer, it is important to not introduce a variable to represent its position.

In the initial state, all four variables farmer, goat, grass, wolf have the value L.

farmer, goat, grass, wolf = L, L, L, L

In the final state, all four variables should have the value R.

farmer, goat, grass, wolf = R, R, R, R

The specification of the problem is

cross_river

-- **inputs: farmer, goat, grass, wolf = L, L, L, L**

-- **outputs: farmer, goat, grass, wolf = R, R, R, R**

subject to the two constraints that

1. the goat cannot be left alone with the grass:

   if goat = grass then farmer = goat

2. the goat cannot be left alone with the wolf:

   if goat = wolf then farmer = goat

6.6.2 Assignment statement

Variables are named boxes to store values. Assignment statement is used to store a value in a variable. It is written with the variable on the left side of the assignment operator and a value on the right side.

variable := value

When this assignment is executed, the value on the right side is stored in the variable on the left side. The assignment

m := 2

stores value 2 in variable m.

m

2

If the variable already has a value stored in it, assignment changes its value to the value on the right side. The old value of the variable is lost.

The right side of an assignment can be an expression.

variable := expression

In this case, the expression is evaluated and the value of the expression is stored in the variable. If the variable occurs in the expression, the current value of the variable is used in evaluating the expression, and then the variable is updated. For example, the assignment

m := m + 3

evaluates the expression m + 3 using the current value of m.

m + 3

= 2 + 3